

Inverse Problems in Geophysics

Part 14: Summary and Outlook

2. MGPY+MGIN

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Summary

- error-weighted discrete (data & model) minimization
- linear Inverse problems
 - only few geophysical problems: gravity, magnetics
 - over- vs. under- vs. mixed-determined problems
 - mostly mixed and ill-conditioned
- non-linear inverse problems: most
 - linearization and use of linear methods

Least-squares vs. minimum-norm solution

$$\mathbf{m}_{LS} = (\mathbf{G}^\top \mathbf{G})^{-1} \mathbf{G}^\top \mathbf{d}$$

- inverse in model space
- finds trade-off between data (minimum squared misfit)

$$\mathbf{m}_{MN} = \mathbf{G}^\top (\mathbf{G} \mathbf{G}^\top)^{-1} \mathbf{d}$$

- inversion in data space
- finds smallest model (0th order regularization)

Singular Value Decomposition

- decomposes problem (matrix) into Eigenvectors/values
- SVD (pseudo) inverse (p_{inv}) as general solution includes LS and MN
- small eigenvalues amplify noise truncated

Regularization

$$\Phi = \Phi_d + \lambda \Phi_m \rightarrow \min$$

- explicit minimization of model objective function
- Occams razor: find simplest model that fits the data
- many ways to define (non)simplicity (e.g. length)
- most common: smoothness (1st/2nd derivatives, geostatistics)
- implementation of prior knowledge (interfaces, point data)

Resolution matrices

- model resolution matrix $\mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G}$
 - point-spread functions for individual parameters
 - diagonal: general resolvability (e.g. resolution radius)
- data resolution $\mathbf{R}^D = \mathbf{G} \mathbf{G}^\dagger$: data correlation
 - importance of individual data $\Rightarrow$ experimental design
- model covariance matrix: correlation and parameter variances

Nonlinear Inversion

- grid search illustrative but too expensive
- linearization $\Rightarrow$ linear subproblem for model update
- iterative solution (improvement) from starting model (important)
- regularization of model (global) or update (local)
- problem of local minimum (strong non-linearity)
- $\Rightarrow$ global methods $\Rightarrow$ model suites

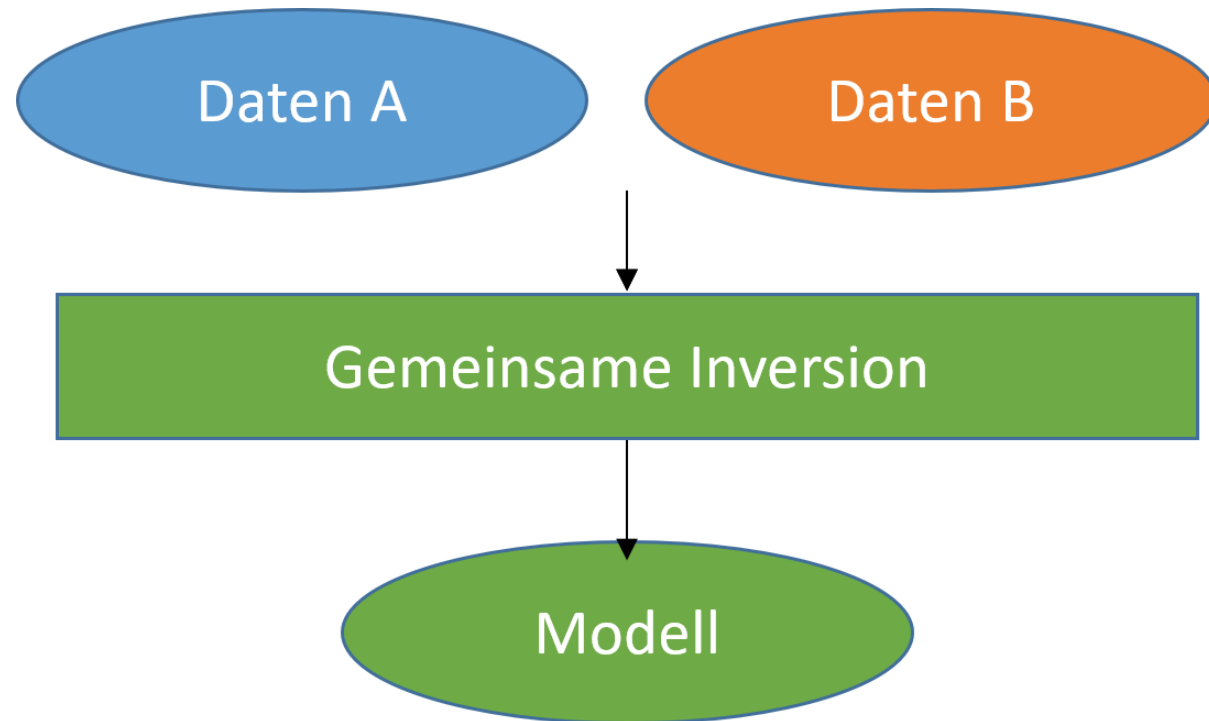
Minimization methods

- gradient method $\nabla_m \Phi$
- non-linear conjugate gradient (NLCG)
- Gauss-Newton minimization
 - explicit computation of Jacobian $\mathbf{J}$
 - implicit Jacobian action $\mathbf{J}\mathbf{x}$ & $\mathbf{J}^\top \mathbf{y}$
- quasi-Newton methods: approximation of Hessian by gradients
 - Limited-Memory quasi-Newton
 - BFGS (Broyden, Fletcher, Goldfarb, and Shanno) algorithm

Global methods

- Bayes theorem is connected to regularized inversion:
 - prior distribution is regularization
 - likelihood describes forward problem and errors
 - posterior solution is trade-off between both
- Monte-Carlo family: MCMC algorithms
- simulated annealing & bio-inspired algorithms: GA, PSO

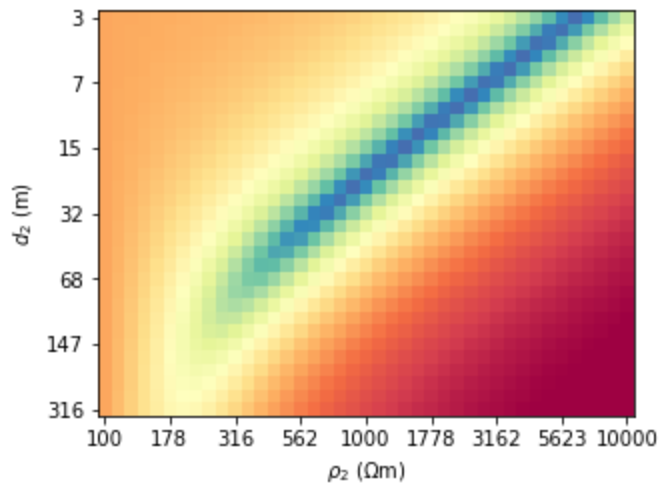
Joint inversion



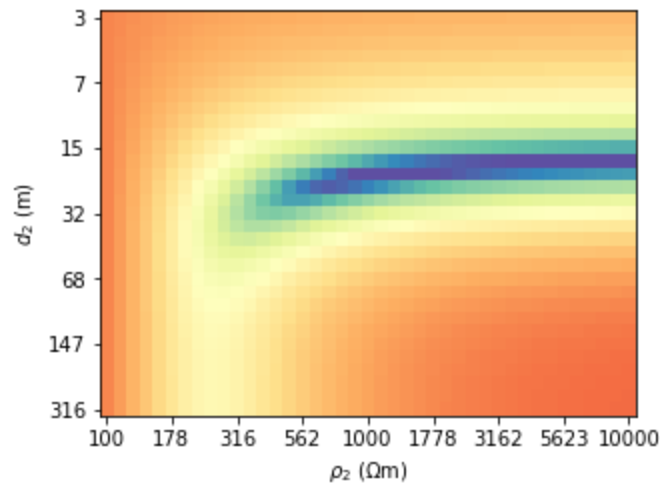
- joint inversion of 2+ data sets
- common model (e.g. conductivity from ERT & EM)
- joint data vector $\mathbf{d} = [\mathbf{d}_1, \mathbf{d}_2]$
- weighting by error $\epsilon = [\epsilon_1, \epsilon_2]$
- improving resolution properties
- improved petrophysics & interpretation

Example joint VES & airborne FDEM

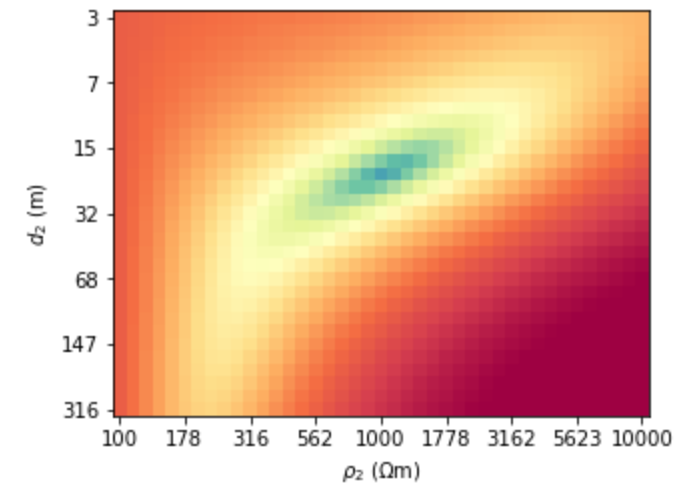
- 20m thick & deep bad conductor ($10000\Omega\text{m}$) embedded in a more conductive ($100\Omega\text{m}$) background
- vertical electrical sounding (VES) and helicopter EM



Φ_d VES

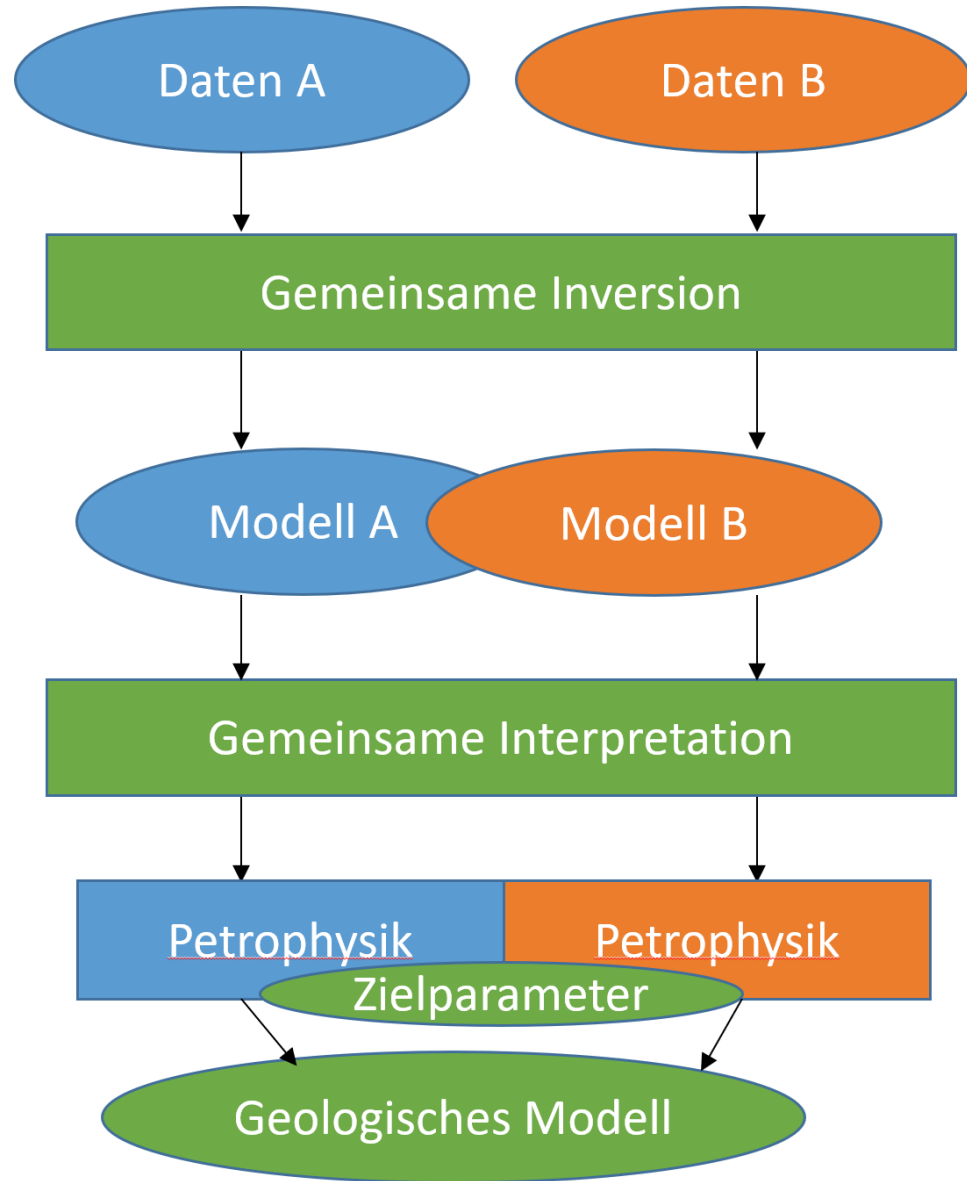


Φ_d HEM



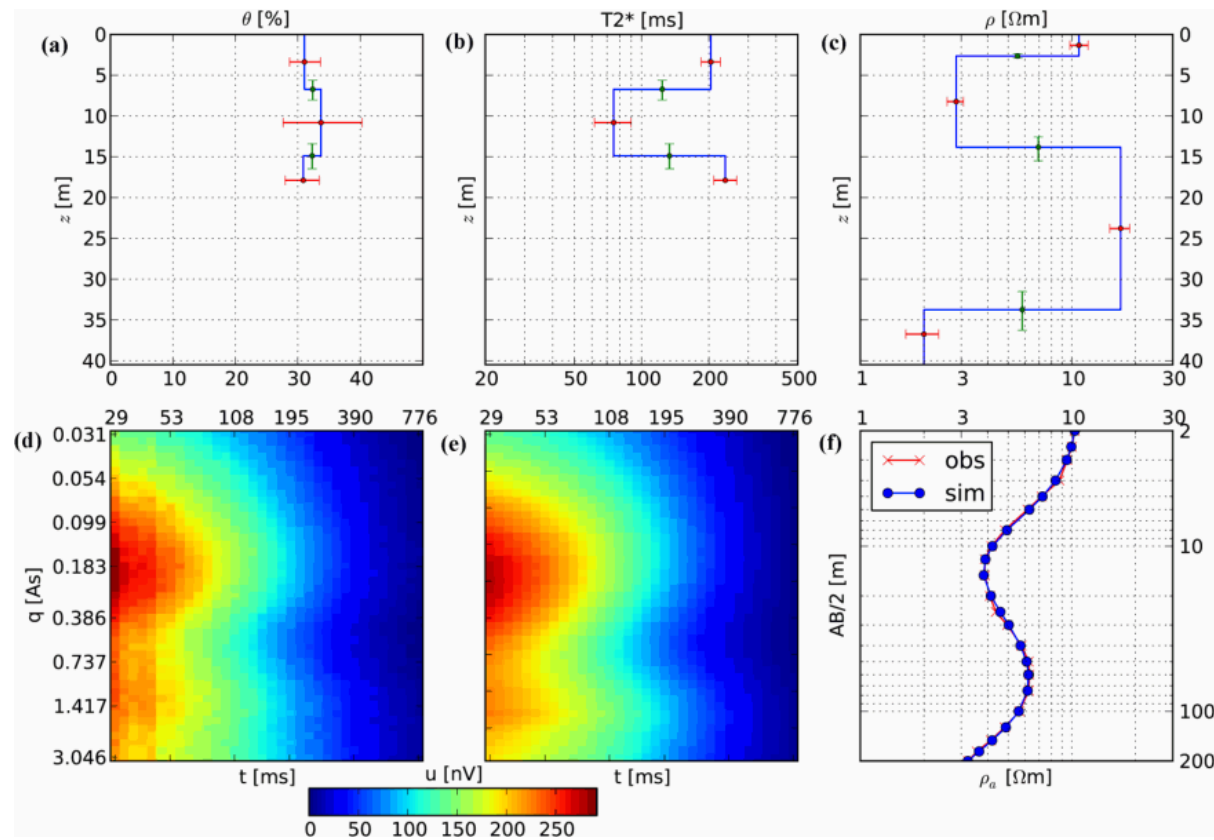
Φ_d joint

Coupled inversion

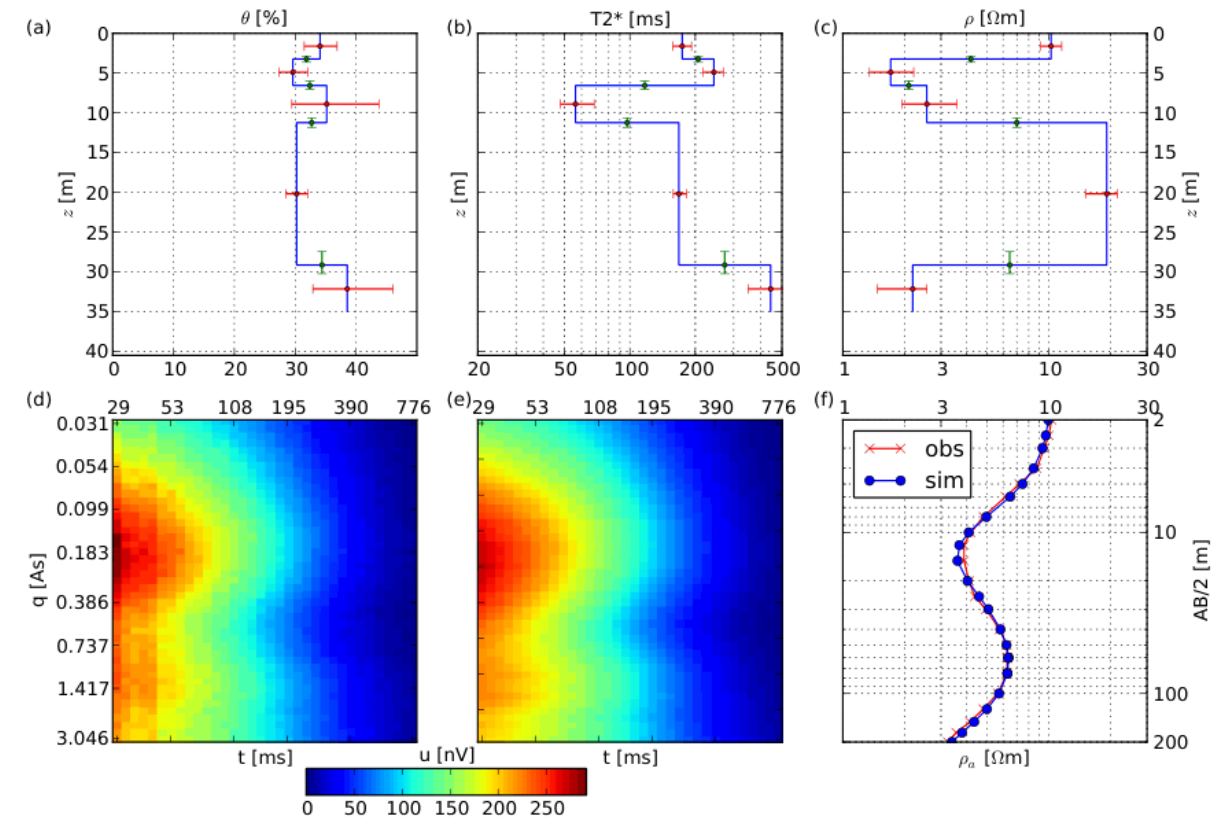


- (structural) coupling of 2+ inversions
- amplification of similarities
- 2+ separate models
- petrophysical conversion $\Rightarrow$ target parameter (porosity, saturation, salinity)
- Common Earth Model (CEM)

Combination MRS & VES

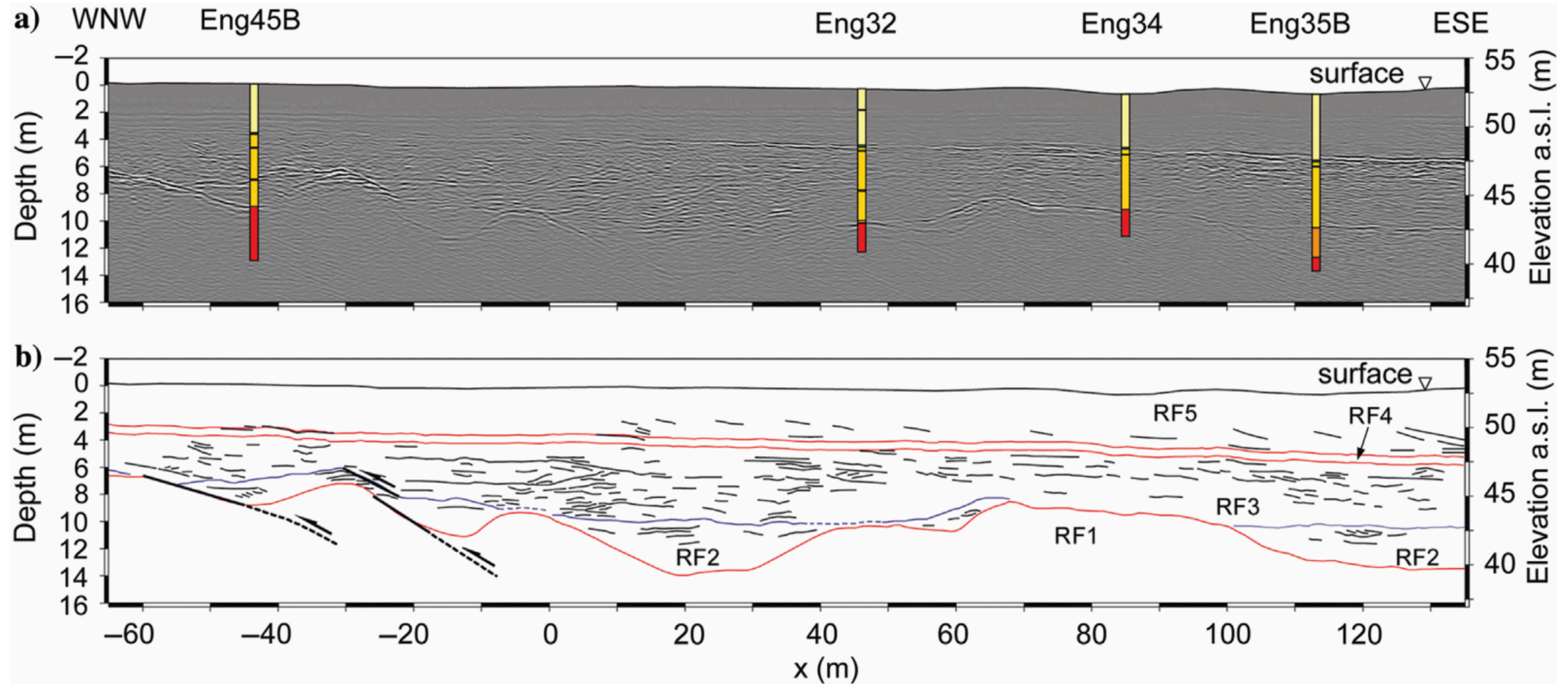


Independent inversion



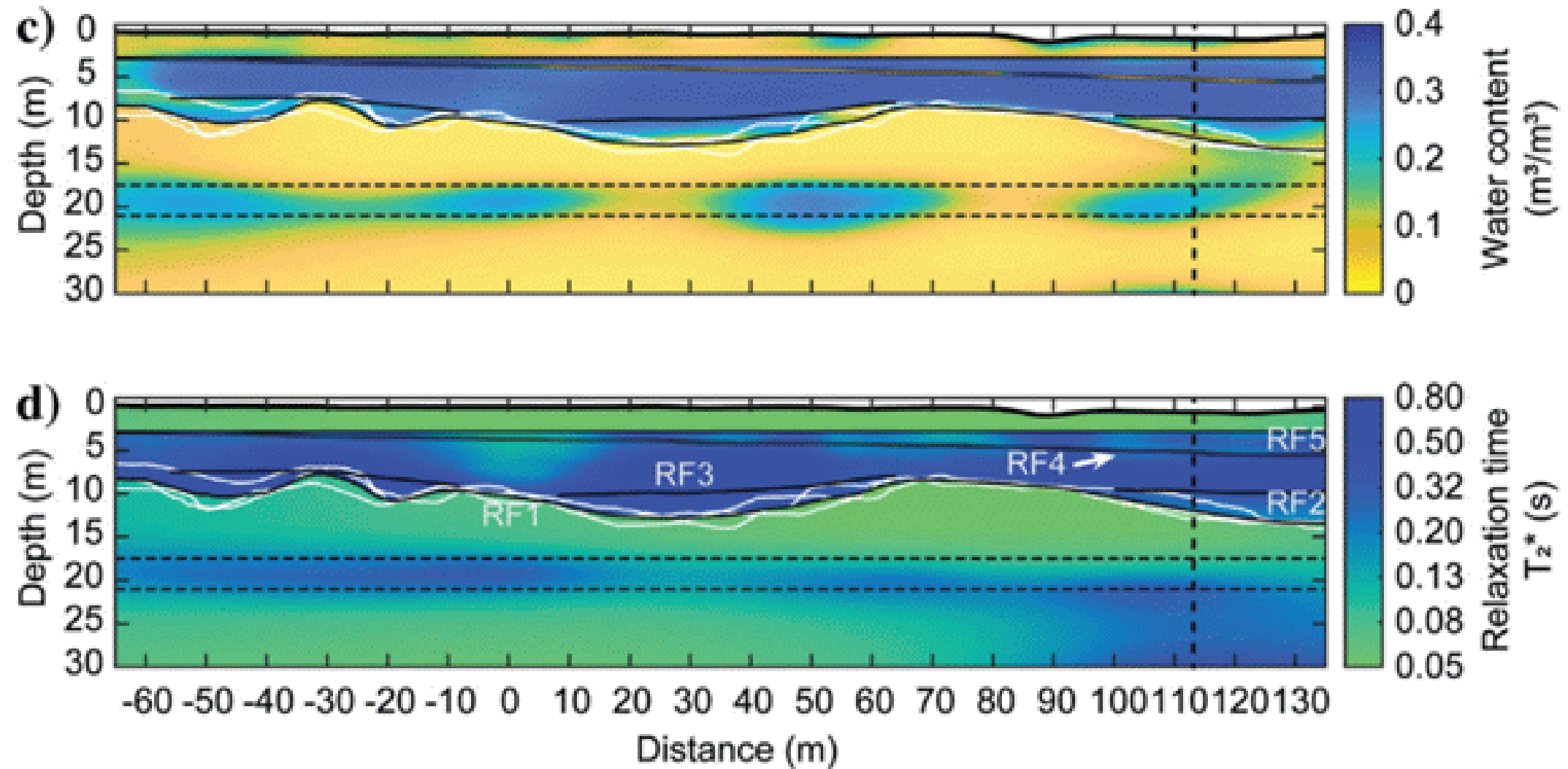
Joint Inversion (G. & Müller-Petke 2012)

Combination GPR & MRT



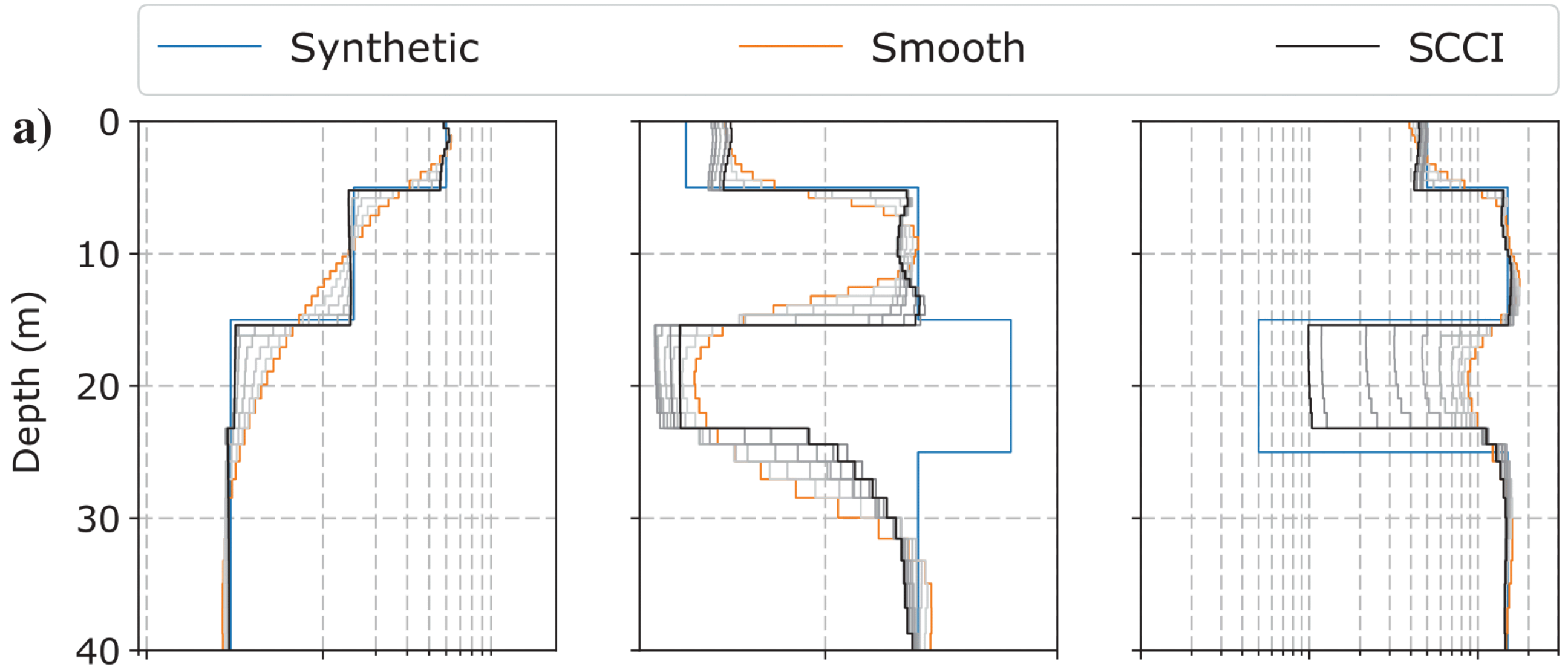
Jiang et al. (2020): Radargram and interpretation

Combination GPR & MRT



Jiang et al. (2020)

Structurally coupled cooperative inversion



Cross-gradient minimization

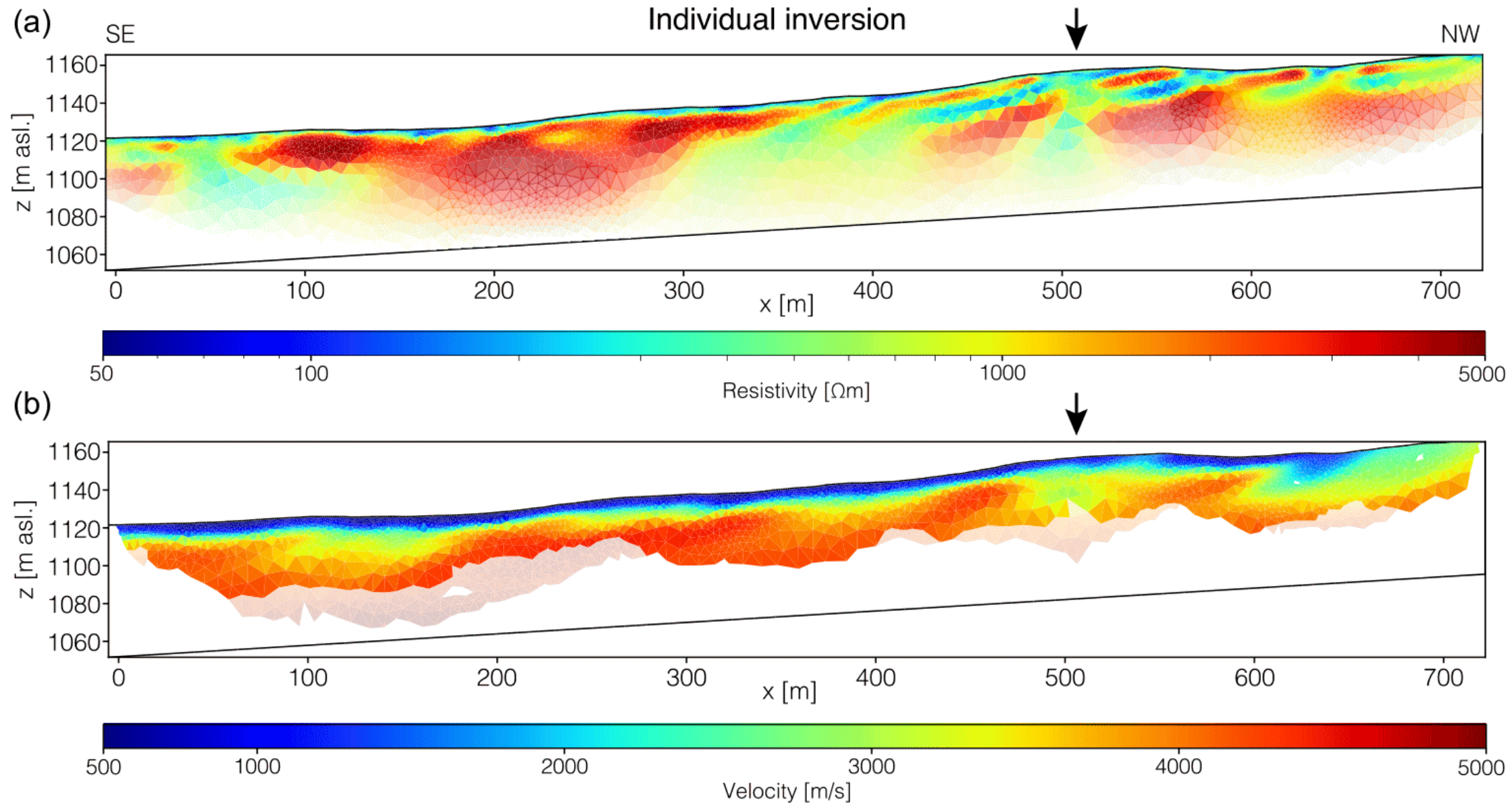
additional minimization of

$$\Phi_{CG} = \lambda_{CG} \nabla \mathbf{m}_1 \times \nabla \mathbf{m}_2$$

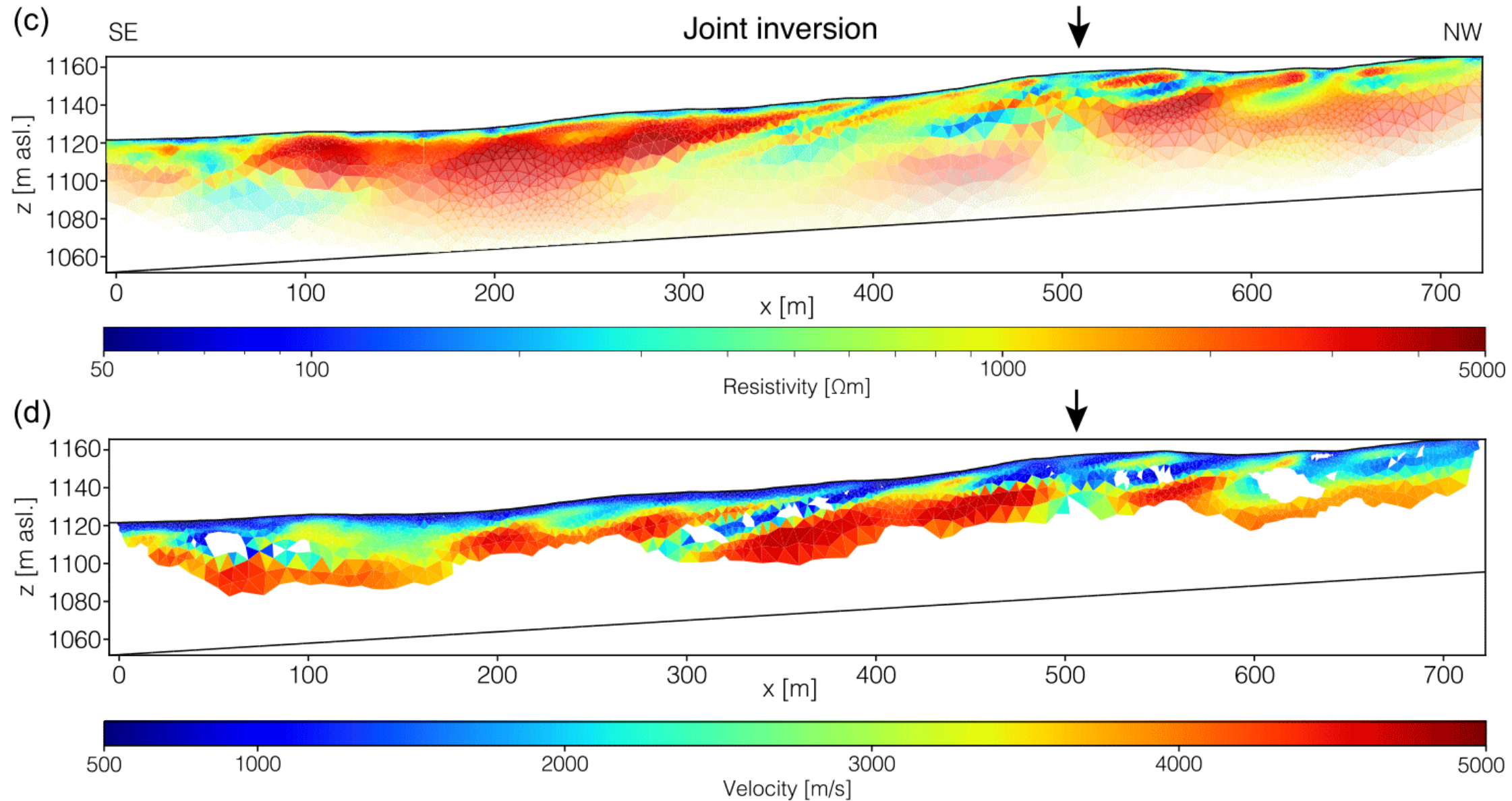
tries to produce model gradients that are either

- parallel (or antiparallel) to each other
- or zero (constant model) in one of the models

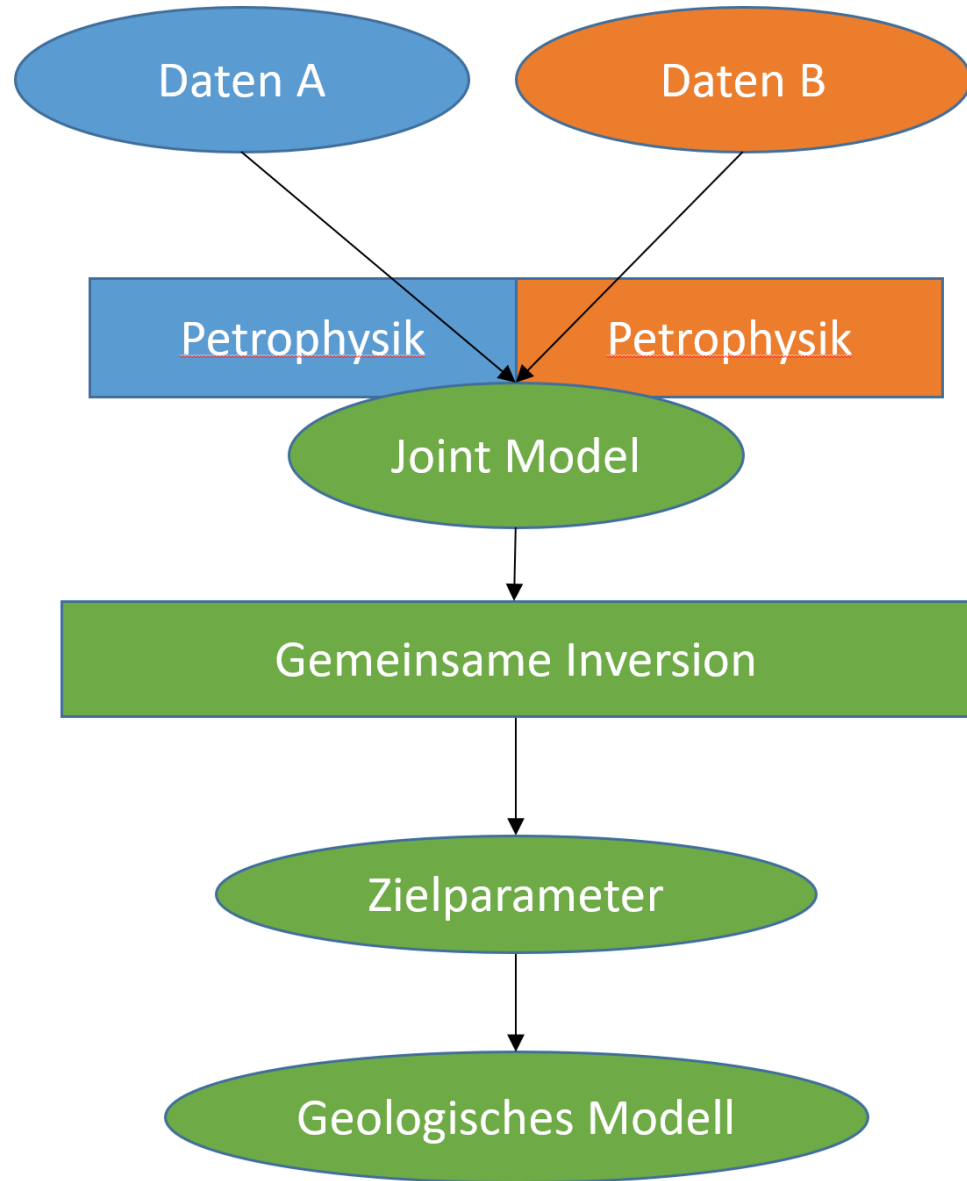
Example from ERT and SRT



Example from ERT and SRT

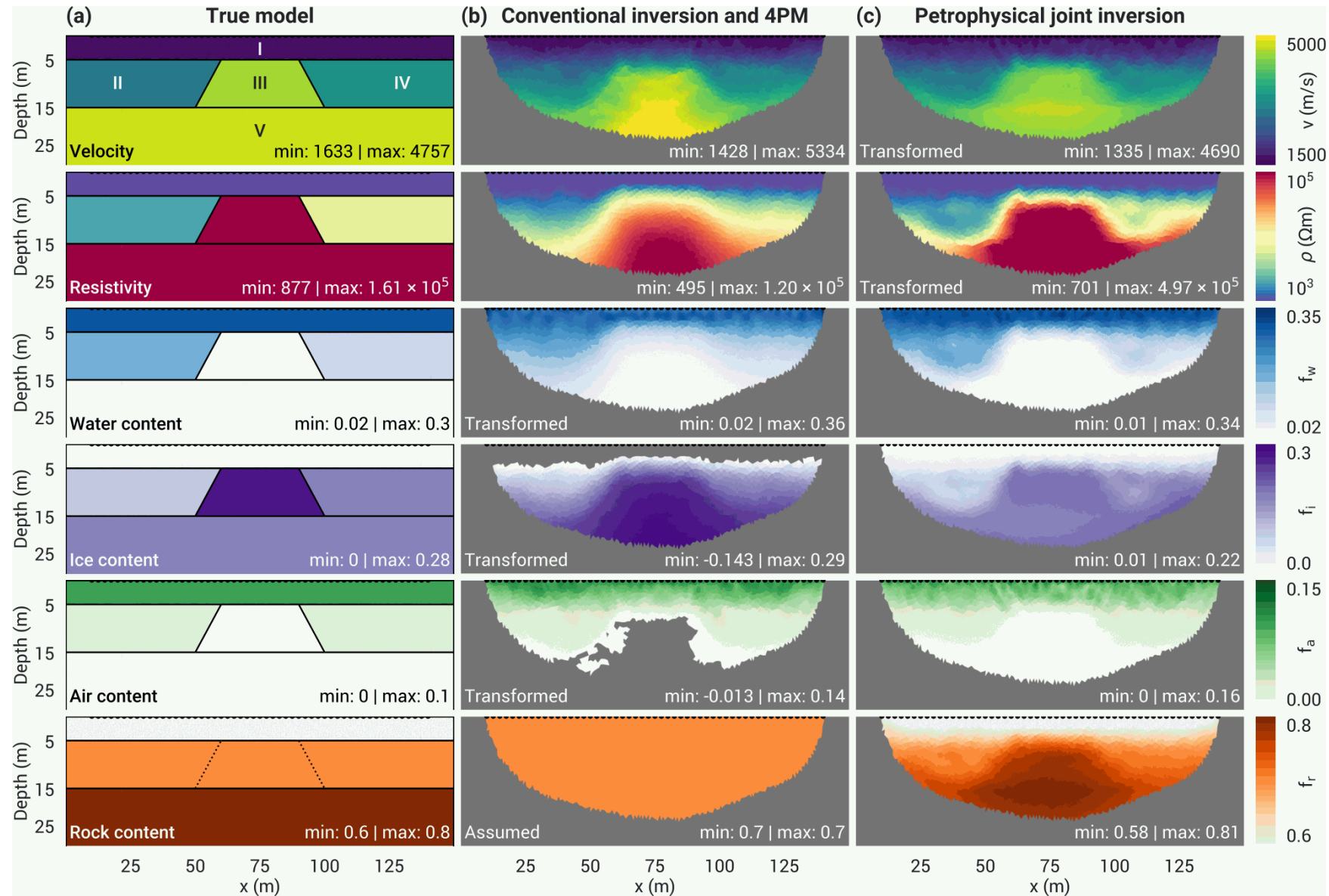


Petrophysical joint inversion



- simultaneous inversion of 2+ data sets
- different physical parameters
- implement petrophysics in inversion
- improving resolution properties
- improved petrophysics & interpretation

Petrophysical JI (Wagner et al. 2019)



Summary

- impression of possibilities but also limits of geophysical inversion
- probabilistic eye on inverse problems
- inversion is trade-off between data fit & assumptions
- knowledge of data errors is crucial (look at misfit!)
- model (& data) transformations helps to constrain

i Regularization is key to subsurface imaging

Add as much information as you have to inversion